

STAT452: Logistic Regression and Classification — Lesson 4 Notes

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1 Introduction to Classification

1.1 The Classification Problem

- **Classification** is the task of predicting a **qualitative (categorical)** response variable Y based on predictor variables $X_1, X_2, \dots, X_p$.
- The goal is to build a function or rule set that takes predictor values as input and predicts the category of Y for previously unseen records.
- Qualitative variables take values in an **unordered set** $\mathcal{C}$, e.g.:
 - Eye color $\in \{\text{brown, blue, green}\}$
 - Insurance claim $\in \{\text{fraudulent, legitimate}\}$
 - Default $\in \{\text{Yes, No}\}$
- Data is typically divided into a **training set** (to build the model) and a **test set** (to validate it).

2 Why Not Linear Regression for Binary Outcomes?

2.1 Encoding and the Problem

For a binary response $Y \in \{0, 1\}$, we can encode $Y = 0$ (No) and $Y = 1$ (Yes) and fit a linear regression:

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

Since $E(Y_i) = P(Y = 1) \cdot 1 + P(Y = 0) \cdot 0 = P(Y = 1)$, this implies:

$$P(\text{Default}) = \beta_0 + \beta_1 x_i$$

2.2 Fundamental Limitation

- Linear regression can produce **predicted probabilities less than 0 or greater than 1**, which is meaningless for probability estimation.
- The linear model does not respect the constraint that probabilities must lie in $[0, 1]$.
- This motivates the use of the **logistic function**, which naturally maps any real-valued input to the interval $(0, 1)$.

3 The Logistic Regression Model

3.1 The Logistic Function

The logistic (sigmoid) function maps any real number to the interval $(0, 1)$:

$$\sigma(x) = \frac{e^x}{1 + e^x} = \frac{1}{1 + e^{-x}}$$

3.2 Model Specification

- **Log-odds (Logit) form:**

$$\log\left(\frac{\pi(\mathbf{X})}{1 - \pi(\mathbf{X})}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = \mathbf{X}^T \boldsymbol{\beta}$$

where $\pi(\mathbf{X}) = P(Y = 1 \mid \mathbf{X})$.

- **Probability form:**

$$\pi(\mathbf{X}) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}} = \frac{e^{\mathbf{X}^T \boldsymbol{\beta}}}{1 + e^{\mathbf{X}^T \boldsymbol{\beta}}}$$

3.3 Odds and Log-Odds

- **Odds:** The ratio of the probability of success to the probability of failure:

$$\text{Odds} = \frac{\pi}{1 - \pi} = \exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)$$

- If $\pi = 0.2$, then Odds = $0.2/0.8 = 0.25$ (for every 1 person who defaults, 4 do not).
- Odds > 1 : success is more likely; Odds < 1 : failure is more likely; Odds = 1: equal chance.
- **Log-odds (Logit):** Taking the log of the odds yields a linear function of the predictors:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

4 Parameter Estimation: Maximum Likelihood

4.1 Likelihood Function

For a sample of size n with binary outcomes $y_i \in \{0, 1\}$:

$$L(\beta; \mathbf{y}, \mathbf{X}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} = \prod_{i=1}^n \left(\frac{e^{\mathbf{X}_i^T \beta}}{1 + e^{\mathbf{X}_i^T \beta}} \right)^{y_i} \left(\frac{1}{1 + e^{\mathbf{X}_i^T \beta}} \right)^{1-y_i}$$

4.2 Log-Likelihood

Taking the log:

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i)] = \sum_{i=1}^n \left[y_i \mathbf{X}_i^T \beta - \log(1 + e^{\mathbf{X}_i^T \beta}) \right]$$

- Maximizing the log-likelihood has **no closed-form solution**.
- Numerical optimization techniques such as **Iteratively Reweighted Least Squares (IRLS)** are used to find $\hat{\beta}$.

5 Interpreting Coefficients

5.1 Odds Ratio Interpretation

- Each coefficient β_j represents the change in **log-odds** for a one-unit increase in X_j , holding other predictors constant.
- Exponentiating gives the **odds ratio**: $\text{OR}_j = e^{\beta_j}$.
- **Interpretation of the odds ratio**:
 - $\text{OR} = 1$: no association between predictor and response.
 - $\text{OR} > 1$: higher predictor values **increase** the odds of success.
 - $\text{OR} < 1$: higher predictor values **decrease** the odds of success.
- A positive β_j increases the log-odds $\Rightarrow$ higher probability of success.
- A negative β_j decreases the log-odds $\Rightarrow$ lower probability of success.

5.2 Example: Default Dataset

With the fitted model $\log\left(\frac{\pi}{1-\pi}\right) = -10.6513 + 0.0055 \cdot \text{balance}$:

- $\hat{\beta}_1 = 0.0055$, so $\text{OR} = e^{0.0055} \approx 1.0055$.
- For each additional unit increase in balance, the odds of default increase by approximately 0.55%.
- At balance = 1000: $\hat{p} = \frac{e^{-10.6513+0.0055 \times 1000}}{1+e^{-10.6513+0.0055 \times 1000}} \approx 0.00576$.
- At balance = 2000: $\hat{p} \approx 0.586$.

5.3 Categorical Predictors

For a dummy-coded predictor (e.g., student: 1 = Yes, 0 = No):

$$\log\left(\frac{\pi}{1-\pi}\right) = -3.5041 + 0.4049 \cdot \text{student}$$

- $\hat{p}(\text{default} \mid \text{student}=\text{Yes}) = \frac{e^{-3.5041+0.4049}}{1+e^{-3.5041+0.4049}} \approx 0.0431$
- $\hat{p}(\text{default} \mid \text{student}=\text{No}) = \frac{e^{-3.5041}}{1+e^{-3.5041}} \approx 0.0292$

6 Statistical Inference

6.1 Wald Test for Individual Coefficients

To test $H_0 : \beta_i = 0$:

$$Z = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}$$

- Z is compared to the **standard normal distribution** to compute the p -value.
- This is analogous to the t -test in linear regression.
- **Confidence interval:** $\hat{\beta}_i \pm z_{1-\alpha/2} \cdot \text{SE}(\hat{\beta}_i)$.

6.2 Likelihood Ratio Test (LRT)

To compare two **nested models**:

$$G^2 = -2 \cdot (\ell_{\text{null}} - \ell_{\text{model}}) \sim \chi_{df}^2$$

- Tests whether adding predictors **significantly improves** model fit.
- df = difference in the number of parameters between the two models.

7 Model Evaluation

7.1 Deviance

$$\text{Deviance} = -2 \cdot \ell(\text{model}) + 2 \cdot \ell(\text{saturated model})$$

- A lower deviance indicates a better fit.
- The deviance of the **null model** (intercept-only) is used as a baseline.

7.2 AIC and Pseudo- R^2

- **AIC:**

$$\text{AIC} = -2\ell(\hat{\beta}) + 2p$$

Balances model fit and complexity. Lower AIC indicates a better model. Can compare **non-nested** models.

- **McFadden's Pseudo- R^2 :**

$$R^2 = 1 - \frac{\ell_{\text{model}}}{\ell_{\text{null}}}$$

Measures improvement over the null model. Interpretation is less direct than R^2 in linear regression.

7.3 Model Selection: AIC vs. LRT

Criterion	Characteristics	Best Use
AIC	Balances fit and complexity; can compare non-nested models	Prediction-focused model selection
LRT	Formal hypothesis test for nested models; based on χ^2 distribution	Significance-focused model comparison

Guiding principle: Use LRT for significance, AIC for prediction. When they disagree (AIC prefers complex model, LRT is not significant), prefer the simpler model for interpretability.

8 Classification Using Logistic Regression

8.1 Decision Rule

Logistic regression predicts the probability $\pi(\mathbf{X}) = P(Y = 1 \mid \mathbf{X})$. To classify:

$$\hat{Y} = \begin{cases} 1 & \text{if } \pi(\mathbf{X}) \geq \tau \\ 0 & \text{if } \pi(\mathbf{X}) < \tau \end{cases}$$

where τ is the **classification threshold** (default $\tau = 0.5$).

The decision boundary $\pi(\mathbf{X}) = 0.5$ corresponds to:

$$\log\left(\frac{\pi}{1-\pi}\right) = 0 \implies \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$$

This is a **linear decision boundary** in predictor space.

8.2 Confusion Matrix

A table showing predicted vs. actual classifications:

	Predicted: 1	Predicted: 0
Actual: 1	True Positive (TP)	False Negative (FN)
Actual: 0	False Positive (FP)	True Negative (TN)

Goal: Maximize TP and TN; minimize FP and FN. Different applications value different types of errors (e.g., fraud detection prioritizes high sensitivity).

8.3 Classification Metrics

- **Accuracy:**

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **True Positive Rate (Sensitivity / Recall):**

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **False Positive Rate:**

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

- **Precision:**

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **F1 Score** (Harmonic mean of Precision and Recall):

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

8.4 Threshold Trade-Off

- **Lowering** τ (e.g., from 0.5 to 0.3): increases True Positives (higher sensitivity) but also increases False Positives (lower specificity).
- **Raising** τ : increases specificity but decreases sensitivity.
- The optimal threshold depends on the **application context** and the relative cost of false positives vs. false negatives.

9 ROC Curve and AUC

9.1 ROC Curve (Receiver Operating Characteristic)

- The ROC curve plots **TPR (sensitivity)** vs. **FPR (1 – specificity)** across all possible thresholds τ .
- The closer the curve to the **top-left corner**, the better the model.
- A **perfect classifier** hugs the top-left corner (TPR = 1, FPR = 0).
- A **random classifier** lies along the diagonal (TPR = FPR).

9.2 AUC (Area Under the ROC Curve)

- AUC measures the model's overall ability to discriminate between classes.
- AUC = 1: perfect model.
- AUC = 0.5: no discrimination (equivalent to random guessing).
- AUC > 0.8: strong discrimination.

9.3 Choosing the Optimal Threshold

Two common strategies:

1. **Youden's Index:** Choose τ to maximize:

$$J = \text{TPR} - \text{FPR}$$

This maximizes the **vertical distance** from the diagonal (random guessing line).

2. **F1-based:** Choose τ that maximizes the F_1 score, balancing Precision and Recall.

10 Summary of Key Formulas

Concept	Formula
Logistic model (logit)	$\log\left(\frac{\pi}{1-\pi}\right) = \mathbf{X}^T \boldsymbol{\beta}$
Probability	$\pi(\mathbf{X}) = \frac{e^{\mathbf{X}^T \boldsymbol{\beta}}}{1 + e^{\mathbf{X}^T \boldsymbol{\beta}}}$
Log-likelihood	$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left[y_i \mathbf{X}_i^T \boldsymbol{\beta} - \log(1 + e^{\mathbf{X}_i^T \boldsymbol{\beta}}) \right]$
Odds ratio	$\text{OR}_j = e^{\beta_j}$
Wald test	$Z = \hat{\beta}_i / \text{SE}(\hat{\beta}_i)$
LRT	$G^2 = -2(\ell_{\text{null}} - \ell_{\text{model}}) \sim \chi_{df}^2$
AIC	$\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}) + 2p$
McFadden's R^2	$R^2 = 1 - \ell_{\text{model}} / \ell_{\text{null}}$
Youden's Index	$J = \text{TPR} - \text{FPR}$